

2024

COMPUTER SCIENCE

Paper : CSMC-103

(Advanced Computer Architecture)

Full Marks : 70

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*

Answer question nos. 1 and 2, and any four questions from the rest.

1. Answer any five questions :

- (a) Define 'instruction execution'.
- (b) Define 'systolic architecture'.
- (c) Differentiate RISC from CISC.
- (d) Derive the formula for average access time in a three-level cache.
- (e) How many states exist in a 4×4 switch, including broadcasting and permutation?
- (f) What are the desirable parameters of ISA?
- (g) Justify the statement – "Algorithm depends on underlying multiprocessor architecture".

$$\textcircled{1} \quad Ht_1 + (1-H)t_2 + (1-H)(1-H)t_3$$

$$\textcircled{2} \quad Ht_1 + (1-H)(t_1+t_2) + (1-H)^2(t_1+t_2+t_3)$$

2. Answer any five questions :

4×5

- (a) Estimate the effect of branch instructions on the performance of pipelined architectures with a suitable set of parameters.
- (b) Split out the design steps for a given latency cycle to get a reservation table. Explain with an example of the latency cycle (2, 3, 2, 5).
- (c) What are the basic assumptions of any message-passing system? Comment on the possible states of any message-passing system.
- (d) Critically comment on the desirable features of a processor, a member of any multiprocessor system. What are the impacts of vector length of vector processors on vectorizing compilers?
- (e) A computer has a 16-way interleaved memory. We are required to access a 64×64 matrix. Compute the total time required for the access if the elements are accessed by (i) row by row (ii) column by column.
- (f) What is a snooping cache? Discuss Write Through and Write Once protocols for Cache consistency.
- (g) Let A be a $2^k \times 2^k$ matrix stored in a row-major order in the main memory. Prove that the transposed matrix A^T can be obtained by performing k-perfect shuffles on A.

Please Turn Over
(0655)

3. (a) Corresponding to the following reservation table, identify the collision matrices. Mention the ORing principle is used to get the modified collision matrices and justify them.

	0	1	2	3	4
S1	A	B		A	B
S2		A		B	
S3	B		AB		A

- (b) For a given reservation table corresponding to pipelined architecture, what will be the upper bound of the Minimal Average Latency (MAL) of incoming jobs? Justify your answer. 5+5
4. (a) Differentiate between Control-flow and Dataflow architectures. What are the alternative forms of Dataflow architectures? Comment on the deficiencies of DFA.
- (b) Draw the data flow graphs to represent the following computation :
- ```

for i ← 1 until m do
 begin
 c(i) ← 0
 for j ← 1 until n do
 c(i) ← a(i, j) x b(j)
 end
 end

```
- (c) What is an instruction template? Give example. 4+4+2
5. (a) Modify the uni-processor matrix multiplication algorithm for two two-dimensional matrices to incorporate temporal and spatial localities. Suggest a suitable systolic array of processors with scheduling for the said problem. Clearly state your assumptions, if any.
- (b) How is the systolic array of processors different from the SIMD environment? Explain briefly. 7+3
6. (a) Write algorithms to add 23 integer values by multiprocessor environments where processors are cube-connected. Estimate the time complexity.
- (b) Compare and contrast the XOR and destination tag-based routing schemes for multistage interconnection networks. 6+4
7. (a) Evaluate the maximum achievable speedup for a computing environment with a large number of processors and programs having 10% sequential part.
- (b) What is the significance of the mismatch between a synthesized and theoretical lower limit of minimal average latency values corresponding to some reservation tables? Justify.
- (c) What are the different types of hazards in pipeline architecture? What are the resolution techniques? Briefly explain. 3+3+4
8. (a) Briefly discuss the alternative flow control strategies in any message-passing system.
- (b) Construct an example showing that the loss of a message can cause a deadlock among a set of communication processes.
- (c) Briefly explain the alternative techniques to reduce cache miss. 3+3+4